

CONTACT

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> Open to relocation

TECHNICAL SKILLS

Robotics & Simulation

ROS / ROS 2, ros2_control, MoveIt, CuRobo, Gazebo, NVIDIA Isaac Sim, Isaac Lab, MuJoCo, URDF, Behavior Trees, Motion Planning

AI / Machine Learning

PyTorch, Reinforcement Learning (SAC/PPO), Sim-to-Real, Domain Randomization, YOLOv8, Computer Vision, Deep Learning, LLM Fine-tuning

Robot Hardware

Denso Cobotta 900, UR5e, Yaskawa GP7, ROX omnidirectional robot, Robotiq 2F-85, OnRobot RG6, Schunk grippers, Intel RealSense D435, Hand-Eye Calibration

Programming

Python, C++, Java

MLOps & Tools

Docker, Git, GitLab, GitHub Actions, CI/CD, Linux, AWS

VIRAAT SAARAN

Robotics Software Engineer | ROS 2 · Motion Planning · Reinforcement Learning · Sim2Real

Robotics Software Engineer with a computer science (AI) background and 2.5+ years at Fraunhofer IPA building ROS 2 manipulation systems end to end (simulation, motion planning, robot modelling and bring-up on Denso Cobotta, UR5e and Yaskawa GP7 arms) and taking the models those systems run on from dataset to deployed robot: rosbag and synthetic dataset creation, YOLOv8 perception, and reinforcement learning policies in NVIDIA Isaac Sim and MuJoCo. Skilled in ROS 2, ros2_control, MoveIt, URDF, Python, PyTorch, RL (SAC/PPO), computer vision, LLM fine-tuning, hand-eye calibration, Docker and CI/CD. Demonstrated a live autonomous pick-and-place cell at Nuremberg Messe 2025; currently completing a thesis on sim-to-real RL for constrained motion planning.

WORK EXPERIENCE

Research Assistant (Wiss. Hilfskraft)

Fraunhofer IPA, Stuttgart, Germany *03/2024 – 08/2026*

- **Simulation & Motion Planning** – Built end-to-end simulation-to-motion-planning workflows in Gazebo, NVIDIA Isaac Sim, MoveIt and CuRobo to generate training data and validate policies before hardware trials; modeled robots in URDF and bridged simulation environments to ROS 2 for repeatable testing.
- **Robot Modeling & Control** – URDF authoring and modification, gripper integration (Robotiq 2F-85, OnRobot RG6, Schunk), topic-based controllers, and hardware interfacing across Denso Cobotta 900, UR5e and Yaskawa GP7 manipulators.
- **Perception & Learning Pipelines** – Developed ROS 2 perception pipelines using YOLOv8 for object detection and pose estimation on Intel RealSense depth streams; created datasets from recorded rosbags, imitation learning, and owned training, evaluation and live inference.
- **Models Running on Hardware** – Deployed trained vision models into running robot cells, closing the loop from model output to grasp execution under real lighting and placement variation.
- **Evaluation & Iteration** – Ran dataset expansion and retraining cycles against failure cases observed on hardware, improving detection robustness under changing lighting and object placement.
- **Robot Cell Deployment** – Supported robot cell setup, hand-eye calibration and frame alignment, hardware integration, and end-to-end testing, including a vision-driven pick-and-place system demonstrated live at Nuremberg Messe 2025.
- **Task Orchestration & MLOps** – Designed Behavior Trees for autonomous task execution, decision-making and failure recovery; containerized ML and robotics workloads with Docker and built CI/CD pipelines (Git, GitLab, GitHub Actions) for reproducible builds, training runs, testing and deployment.

ROS 2, ros2_control, MoveIt, CuRobo, Gazebo, NVIDIA Isaac Sim, URDF, Behavior Trees, Python, PyTorch, YOLOv8, Computer Vision, Dataset Creation, rosbags, Intel RealSense, Hand-Eye Calibration, Denso Cobotta 900, UR5e, Yaskawa GP7, ROX omnidirectional platform, Robotiq 2F-85, OnRobot RG6, Schunk grippers, C++, Docker, CI/CD, Git, GitLab, GitHub Actions, Linux

EDUCATION

M.Sc. Computer Science

University of Stuttgart
04/2023 – 2027 (expected)
Focus: Autonomous Systems, Robotics, ML

B.Tech, Computer Science

Dr. A.P.J. Abdul Kalam Technical University, India
05/2017 – 08/2021

SOFT SKILLS

Problem Solving

Team Work & Collaboration

Leadership

Communication

Self-driven & Quick Learner

Cross-functional Collaboration

LANGUAGES

Hindi – Native

English – Professional Fluency

German – Basic (A2, improving)

ACHIEVEMENTS

2 peer-reviewed research paper publications (IGI Global, Springer)

Smart India Hackathon – National Finalist

Microsoft MTA-98-381 Certified Python Programmer

Master's Thesis — Adaptive RL for Constrained Motion Planning of Robotic Manipulators

Fraunhofer IPA, Stuttgart, Germany *09/2026 – Present*

- Designing and training reinforcement learning policies for constrained, contact-rich pick-and-place inside a confined robotic work cell.
- Training and evaluating policies in NVIDIA Isaac Sim / Isaac Lab and MuJoCo; shaping reward functions and training curricula, and applying domain randomization to improve sample efficiency and policy robustness.
- Researching sim-to-real transfer strategies (system identification, observation noise and latency modeling) to close the reality gap for deployment on a physical manipulator.
- Benchmarking learned policies against classical planners (RRT*, trajectory optimization) and evaluating physics-engine backends (Newton, PhysX) for training throughput.

Reinforcement Learning (SAC/PPO), Sim-to-Real Transfer, Domain Randomization, Reward Shaping, NVIDIA Isaac Sim, Isaac Lab, MuJoCo, Newton, PhysX, Motion Planning, RRT, PyTorch, Python*

Assistant System Engineer

Tata Consultancy Services, India *08/2021 – 03/2023*

- Worked in an agile delivery team as a Database Administrator and later as a Java Developer on a ServiceNow-based enterprise platform.

R&D Machine Learning Intern

E-bibil Technologies Pvt. Ltd., Remote *05/2020 – 09/2020*

- Built an ML model and supporting API for object detection using a YOLO-based algorithm as part of a confidential R&D project.

PROJECTS

LEGO Pose Estimation – Nuremberg Messe

2025 · ROS 2 · Intel RealSense · Behavior Trees · Docker

End-to-end ROS 2 LEGO pose-estimation pipeline (Intel RealSense) integrating perception, Behavior Trees and robot control for autonomous, colour-based pick-and-place. Fully dockerized and run continuously as a public demonstration at Nuremberg Messe 2025.

Dynamic Path Planning

ROS 2 · Gazebo · Isaac Sim · MoveIt · CuRobo

ROS 2-based dynamic path planning system integrating Gazebo, Isaac Sim, MoveIt and CuRobo, using topic-based controllers for simulation-to-real manipulation workflows. Compared planner behaviour across simulators to identify where planned trajectories diverged on real hardware.

MathGPT

2025 · GPT-2 · Cloud GPU Clusters

Fine-tuned a GPT-2-based language model on a mathematical reasoning dataset to study where LLM reasoning breaks down. Handled dataset preparation, distributed fine-tuning on cloud GPU clusters, and benchmarking against the base model.

PUBLICATIONS & CERTIFICATIONS

Reconstruction of 2D Images Into 3D – IGI Global (2021), DOI: 10.4018/978-1-7998-3327-7.ch002

A Literature Review on GANs with Applications in Healthcare – Springer (2020), DOI: 10.1007/978-981-33-6981-8_18